

Infrastructure as Code (IaC) Promotion Pipeline

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Introduction (Use this exactly or with small edits)

Infrastructure as Code (IaC) has become a core practice in modern cloud and DevOps environments, enabling infrastructure to be provisioned, managed, and version-controlled using code. This approach reduces human error, improves consistency across environments, and allows infrastructure changes to follow the same lifecycle as application code.

This project implements an **IaC Promotion Pipeline** using **CloudFormation**, **Terraform**, and **GitHub Actions** to demonstrate how infrastructure changes can be promoted safely from a development environment to a production environment. The solution focuses on automating deployments while enforcing governance controls such as manual approvals and drift detection.

A simple static web application hosted on **Amazon S3** is used as the target infrastructure. The infrastructure is first defined using CloudFormation and then reimplemented using Terraform to highlight tool-based IaC practices. A CI/CD pipeline is built to automatically deploy changes to a development environment and require explicit approval before deploying to production.

Additionally, the project demonstrates **configuration drift detection and correction** by introducing manual changes directly in the AWS console and restoring the intended state through IaC. This ensures that infrastructure remains consistent with its declared configuration and reinforces the principle of code as the single source of truth.

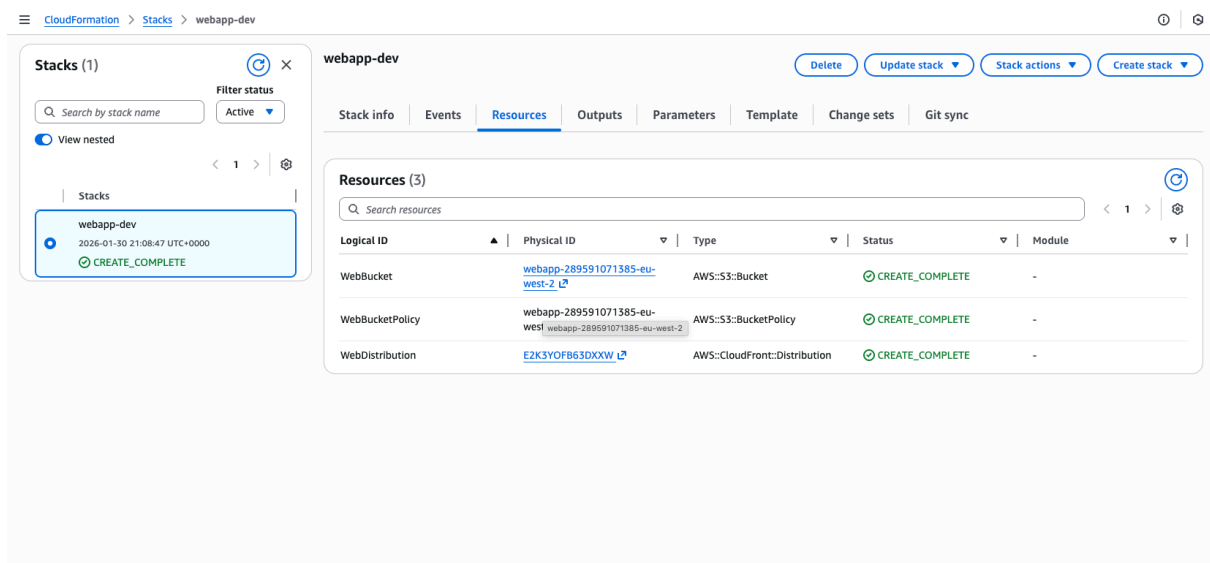
Through this project, key DevOps concepts such as automation, environment promotion, approval gates, and drift management are applied in a practical cloud-based scenario.

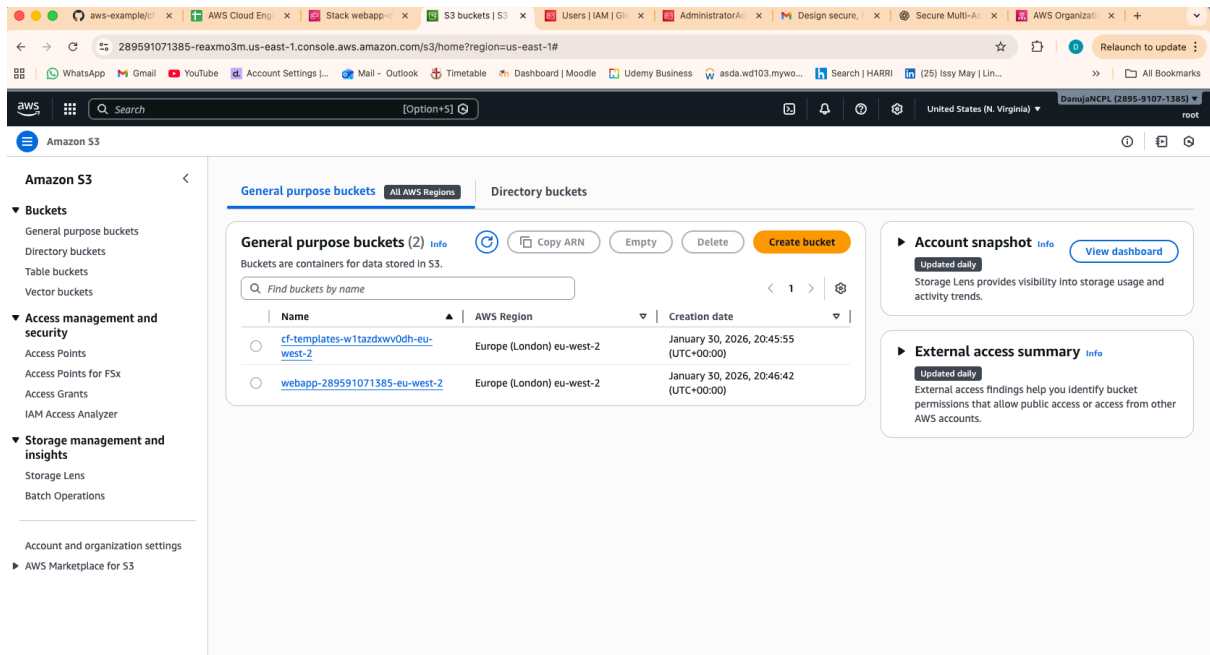
STEP 1 — Create the simple CloudFormation stack

What you are building

A CloudFormation template that creates:

- S3 bucket
- CloudFront distribution
- Output = CloudFront URL

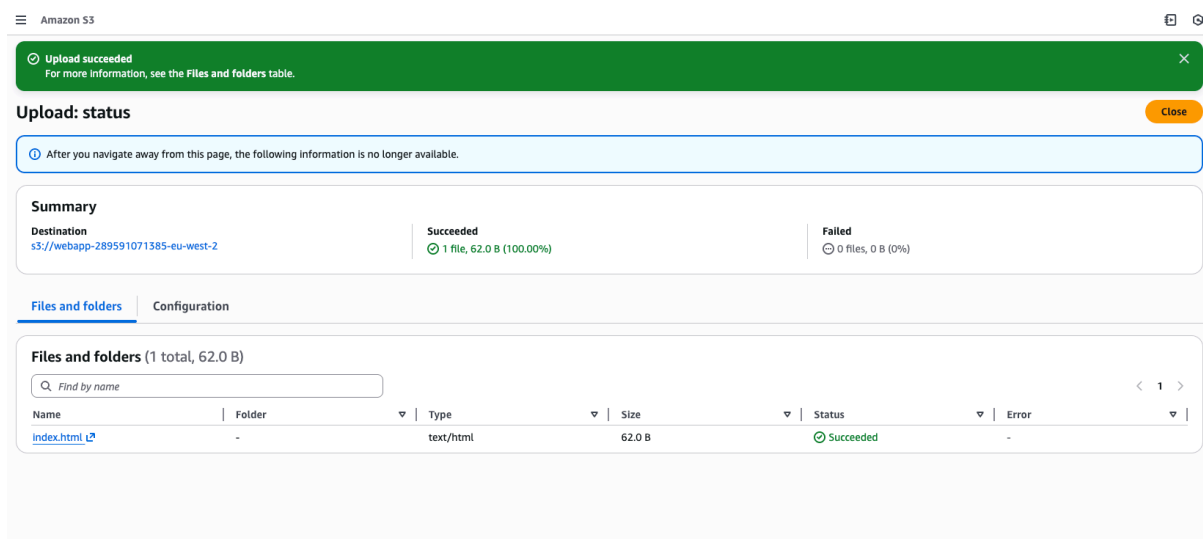




loudFormation → webapp-dev → Resources → click WebBucket

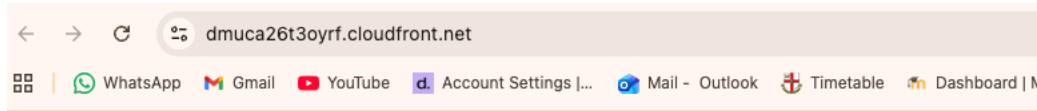
Then:

Upload → add `index.html` → Upload



Step 5 — Test

CloudFormation → Outputs
Copy WebsiteURL
Open in browser

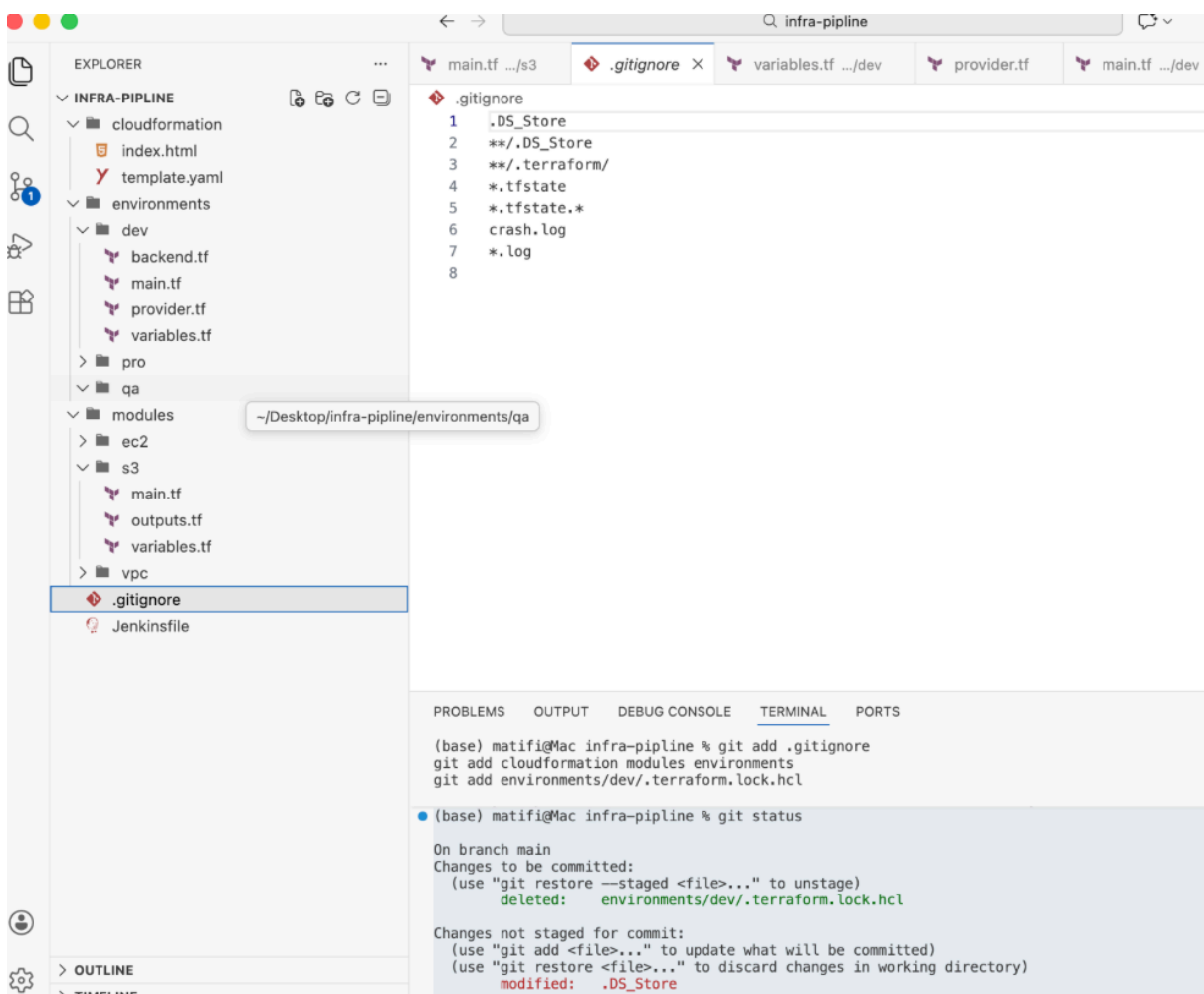
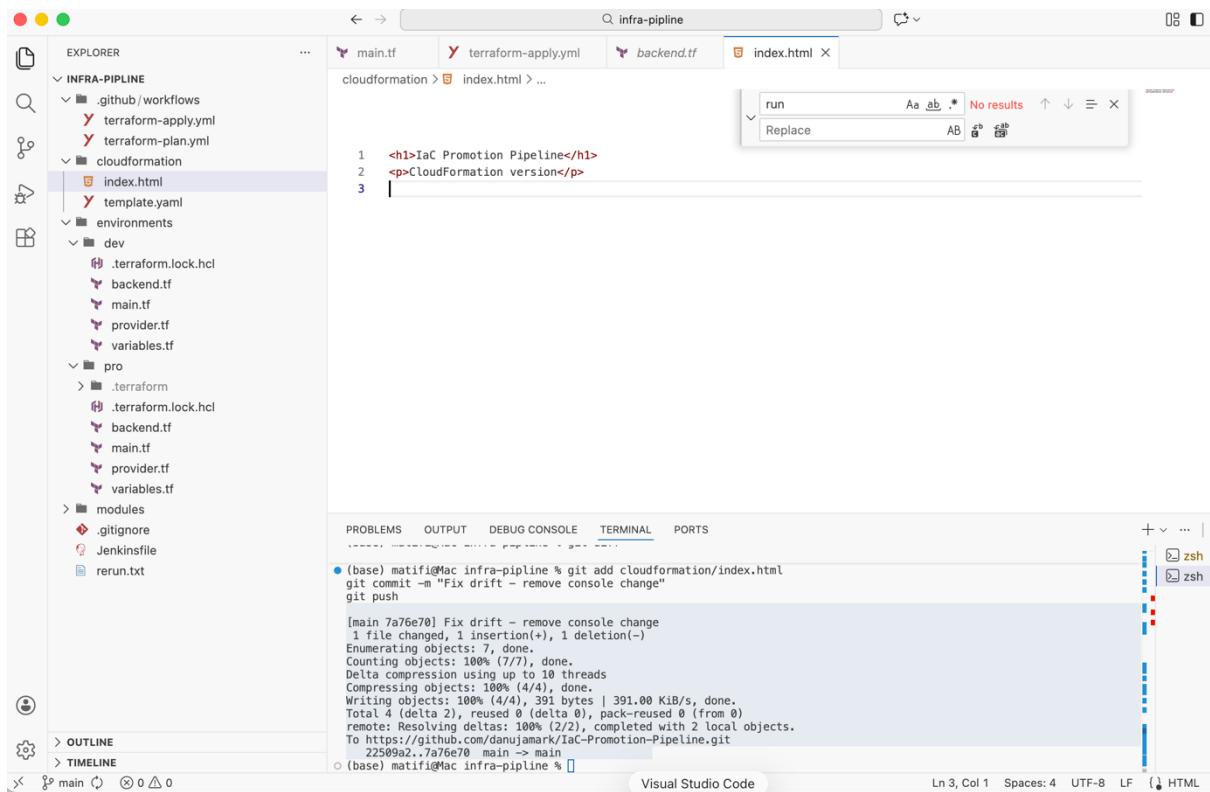


IaC Promotion Pipeline

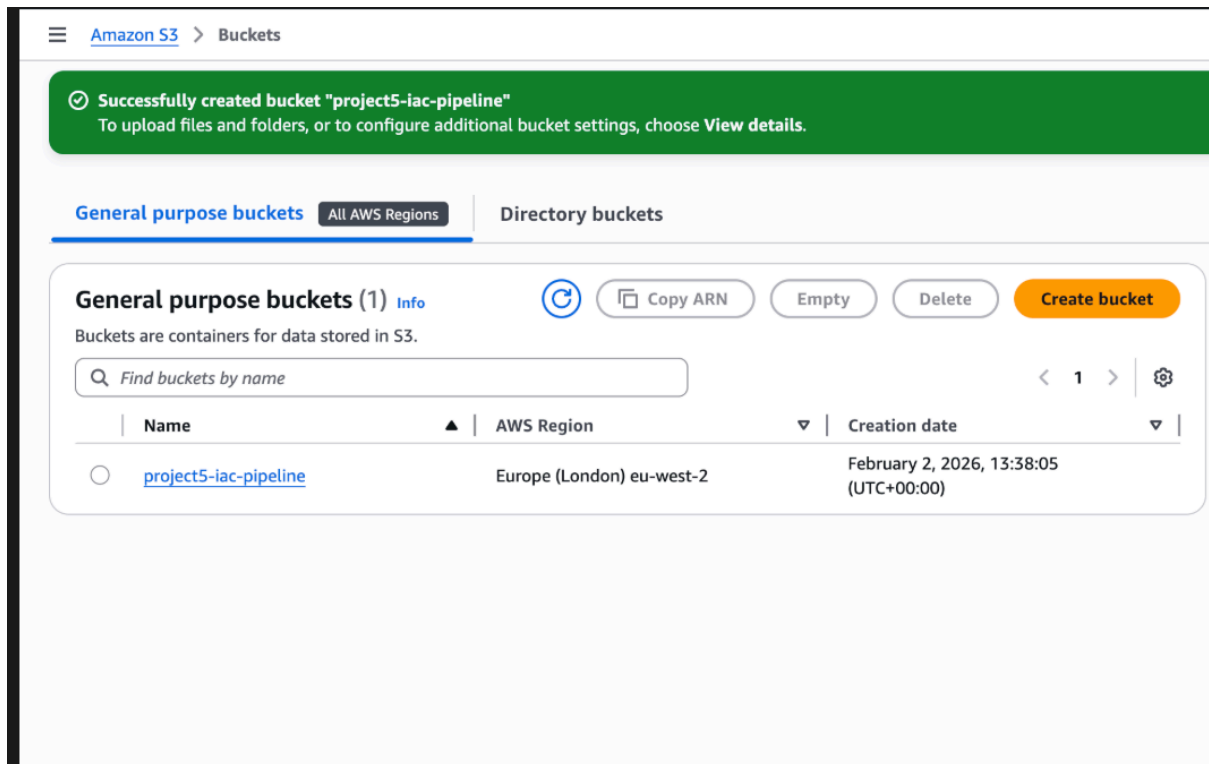
CloudFormation version

Repository Structure

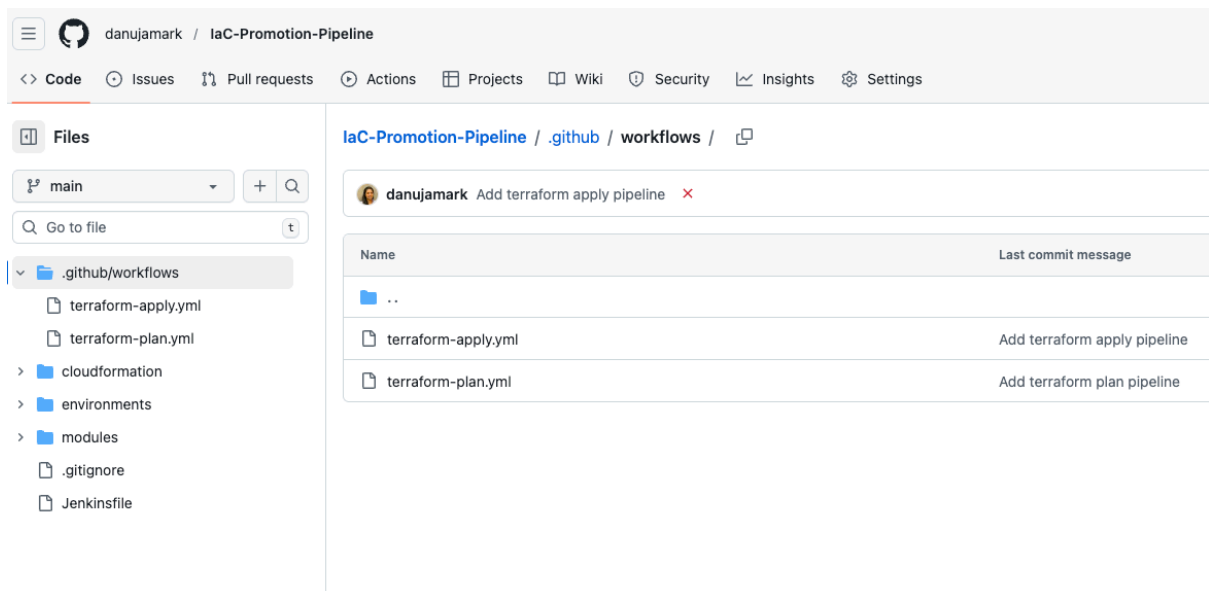
- Project folders:
 - .github/workflows
 - cloudformation
 - modules
 - environments/dev
 - environments/pro
 -



Create the bucket



Verify GitHub



Add secret key to git

- ⚙️ General
- Access
- 👤 Collaborators
- 🗨️ Moderation options
- Code and automation
- 🔗 Branches
- 🏷️ Tags
- 📋 Rules
- 🔄 Actions
- 🔗 Models Preview
- 🔗 Webhooks
- 🤖 Copilot
- 📁 Environments
- 💻 Codespaces
- 📄 Pages
- Security
- 🔒 Advanced Security
- 🔑 Deploy keys
- * Secrets and variables

Actions

Codespaces

Dependabot

Actions secrets and variables

Secrets and variables allow you to manage reusable configuration data. Secrets are **encrypted** and are used for sensitive data. [Learn more about encrypted secrets](#). Variables are shown as plain text and are used for **non-sensitive** data. [Learn more about variables](#).

Anyone with collaborator access to this repository can use these secrets and variables for actions. They are not passed to workflows that are triggered by a pull request from a fork.

SecretsVariables

Environment secrets

This environment has no secrets.

Manage environment secrets

Repository secrets

New repository secret

Name	Last updated
🔒 AWS_ACCESS_KEY_ID	now  
🔒 AWS_SECRET_ACCESS_KEY	now  

Approval Gate

- GitHub Environments → `pro`
- Required reviewers enabled
- Pipeline waiting for approval
- Approved deployment

jamark / IaC-Promotion-Pipeline

issuesPull requestsActionsProjectsWikiSecurityInsightsSettings

General

Access

Collaborators

Moderation options

Code and automation

Branches

Tags

Rules

Actions

Models

Webhooks

Copilot

Environments

Codespaces

Pages

Security

Advanced Security

Deploy keys

Secrets and variables

Integrations

Environments / Configure pro

Deployment protection rules

Configure reviewers, timers, and custom rules that must pass before deployments to this environment can proceed.

☒ Required reviewers

Specify people or teams that may approve workflow runs when they access this environment.

Add up to 5 more reviewers

Search for people or teams...

danujamark

☐ Prevent self-review

☐ Wait timer

Set an amount of time to wait before allowing deployments to proceed.

Enable custom rules with GitHub Apps

Learn about existing apps or create your own protection rules so you can deploy with confidence.

☒ Allow administrators to bypass configured protection rules

Save protection rules

Deployment branches and tags

Limit which branches and tags can deploy to this environment based on rules or naming patterns.

No restriction

Environment secrets

terraform-apply

test approval gate #10

Summary

All jobs

apply-dev

apply-pro

Run details

Usage

Workflow file

Triggered via push 1 minute ago

Status

Total duration

Artifacts

Waiting

danujamark pushed · 22509a2

danujamark requested your review to deploy to pro

Review deployments

terraform-apply.yml

on: push

apply-dev

20s

apply-pro

pro waiting for review

Deployment protection rules

Reviewers, timers, and other rules protecting deployments in this run

Start all waiting jobs

Event	Environments	Comment
danujamark requested review	pro	-

← terraform-apply
✓ trigger pipeline #8

Summary

All jobs

- ✓ apply-dev
- ✓ apply-pro

Run details

- Usage
- Workflow file

Triggered via push 1 minute ago

danujamark pushed → f5d09b5 main

Status

Success

Total duration

54s

Artifacts

—

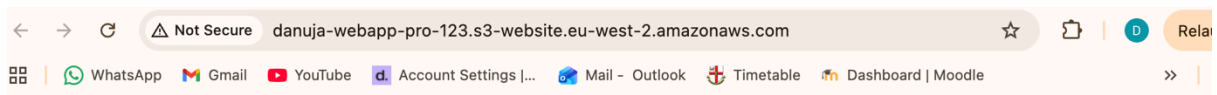
terraform-apply.yml

on: push



Section 6 – Website Output

- Browser showing website
- Version v1
- Version v2



IaC Promotion Pipeline

CloudFormation version

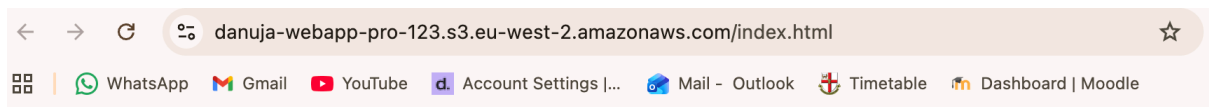


IaC Promotion Pipeline

CloudFormation version changedv 2.0

Drift Demonstration

- Edited `index.html` in S3 console
- Website showing HACKED FROM CONSOLE



IaC Promotion Pipeline

CloudFormation version

HACKED FROM CONSOLE

Section 8 – Drift Fix

- Git commit “Fix drift”
- Pipeline success after fix
- Website showing correct text again

danujamark / IaC-Promotion-Pipeline

CodeIssuesPull requestsActionsProjectsWikiSecurityInsightsSettings

terraform-apply

Fix drift - remove console change #11

Summary

All jobs

apply-devapply-pro

Run details

Usage

Workflow file

Triggered via push 7 hours ago

Status

Total duration

Artifacts

danujamark pushed -> 7a76e70main

Success

2m 9s

-

terraform-apply.yml

on: push

apply-dev20s

apply-pro18s

Deployment protection rules

Reviewers, timers, and other rules protecting deployments in this run

Event	Environments	Comment
danujamark approved 7 hours ago	pro	apro dirt

danuja-webapp x danuja-webapp x danuja-webapp x S3 buckets | S x Console Home x Google x

danuja-webapp-pro-123.s3.eu-west-2.amazonaws.com/index.html

Relaun

WhatsApp Gmail YouTube Account Settings |... Mail - Outlook Timetable Dashboard | Moodle

IaC Promotion Pipeline

CloudFormation version changedv 2.0